

Pollen & Pain: Forecasting Neighborhood-Level Asthma Emergency Department Surges in New York City: **Saketh Boddu, Andre Nguyen, Nam Lai**

Objective:

- This project forecasts monthly asthma emergency department (ED) visit surges across New York City at the Neighborhood Tabulation Area (NTA) level, producing locally actionable predictions for schools, clinics, and public health planners.
- Using historical pollen counts, weather data, urban tree canopy, and asthma ED records as core inputs, and supported by 311 complaints, community health survey data, and Reddit activity.
- The aim is to develop a more localized framework that connects environmental exposure to asthma outcomes at the neighborhood level, where public health decisions are actually made.

Significance:

- Asthma affects millions of New York City residents, and certain high-density, environmentally burdened neighborhoods experience a greater health burden.
- Neighborhood-level monthly predictions can directly inform public health resource allocation, school planning, and clinical preparedness.
- This project is impactful from a data science perspective because it brings together multiple data types, including environmental data, health records, and public response signals, rather than relying on a single source.
- The system can be integrated into other cities with valid data.

Literature Review

- Pollen exposure is a well-documented asthma trigger, with ED visits typically peaking one to two weeks after elevated airborne counts. Weather conditions, urban vegetation, and neighborhood stressors further shape the extent of residents' exposure.
- Recent work has tested whether digital signals, such as social media posts and 311 complaints, can track respiratory health trends in near real time, with mixed but promising results.
- The consistent gap across this literature is geographic resolution. Most studies operate at the city or county level, masking the within-city variation that drives health disparities in a place like New York.
- Prior forecasting tools typically rely on regional pollen averages or single environmental predictors. Few systems integrate environmental exposure, clinical outcomes, and behavioral signals at the neighborhood scale in NYC.
- Asthma burden, tree canopy, housing conditions, and allergen exposure all vary significantly across neighborhoods. However, the forecasting tools available to clinics and schools lack that granularity.
- This project builds on those foundations by combining environmental, clinical, and behavioral data at the NTA level, producing predictions that are actionable at the scale where public health decisions actually get made.

Proposed solution:

- Build a dual-database architecture using PostgreSQL for structured time-series data (pollen, weather, ED visits) and MongoDB for unstructured text and logs (Reddit posts, 311 complaints), with all historical datasets ingested through a one-time ETL script.
- Spatially align all datasets to the NTA level using GeoPandas spatial joins for lat/long sources, a ZIP-to-NTA crosswalk for pollen data, and a UHF-to-NTA crosswalk for

Community Health Survey data, treating CHS as a zone-level control rather than a neighborhood-level feature.

- Apply keyword-based NLP to Reddit r/nyc posts and 311 Air Quality complaints to produce a monthly behavioral allergy signal for model validation.
- Train a Logistic Regression baseline and an XGBoost primary model to predict monthly asthma ED visit rates per NTA, excluding neighborhoods with fewer than 12 monthly cases to avoid unstable small-count predictions.
- Surface results in a Streamlit dashboard with neighborhood-level risk maps and feature-importance visualization, with live pollen forecast data refreshed on page load via the Google Pollen API.

Data Collection and Sources:

Type	Source	Data	Access
Web Scraping	Pollen.com JSON API	Daily pollen index (tree, grass, weed, mold) by NYC ZIP code (30-day history per call).	Free, No Key
API	Google Pollen API	Live 5-day pollen forecast by lat/long (for dashboard display only).	Free Tier
API	Open-Meteo Weather Archive	Historical daily temperatures, precipitation, and wind speed for each neighborhood centroid.	Free, No Key
Bulk Download	NYC Open Data — Asthma ED Visits	Monthly asthma ED visit rates by NTA neighborhood.	Free, Direct
Bulk Download	NYC Street Tree Census	683,788 street trees with NTA codes (aggregated to canopy density score).	Free, Direct
Log	NYC 311 Service Requests	Air Quality complaints 2019–2024, geo-tagged.	Free, Direct
Survey	NYC Community Health Survey	Annual self-reported respiratory health by UHF neighborhood.	Free, Public
Web Scraping	Reddit r/nyc via PRAW	Posts and comments filtered by allergy symptom keywords (used as a qualitative validation signal).	Free, Library

- This project uses public environmental, health, and behavioral datasets, including pollen, weather, asthma ED visits, tree canopy, 311 complaints, community health survey data, and Reddit posts.
- Most of the data is publicly available or accessible via free APIs, downloads, or scraping tools, making the project feasible to implement.
- The datasets use different geographic units, including ZIP codes, NTAs, and UHFs, so they need to be aligned to a common neighborhood-level structure.
- The data also differ in time scale, ranging from daily to annual so that that aggregation will be required before modeling.

- A possible limitation is that historical pollen coverage may be less complete than that of the other datasets, so pollen features will need to be validated before modeling.

Data Storage and Processing Architecture

- **Database Setup:**
 - **Relational Database (PostgreSQL):** Stores structured time-series data (pollen_daily, ed_visits_monthly, weather_controls)
 - **Document Store (MongoDB):** Stores unstructured text and logs (Reddit posts, 311 complaint text, and API response logs)
- **Data Management:** All historical datasets are ingested via a one-time ETL script. The Streamlit dashboard refreshes pollen forecast data on page load via the Google Pollen API.

Machine Learning Model Selection and Implementation

- **Model selection:**
 - **Logistic Regression (Baseline):** Binary classifier predicting above-median asthma ED visit month.
 - **XGBoost (Primary):** Regression model predicting monthly per-capita ED visit rate.
- **Feature engineering:** Includes a Composite Pollen Index (weighted), 2-week and 4-week Pollen Lag Features, Canopy Density Score, Weather Controls, CHS, and Respiratory Baseline.
- **Model Training and Validation:** The model will be trained using the earliest available historical data and evaluated on a later holdout period. Walk-forward time-series cross-validation will be used, with the model trained on past data and evaluated on future observations, mimicking real-world forecasting conditions.

Evaluation and Success Metrics

- **Performance Metrics:**
 - The project uses **MAE** and **RMSE** to evaluate regression performance.
 - **AUC-ROC** is used to evaluate the binary classification baseline model.
 - **Pearson correlation** is used to measure the relationship between predictions and actual outcomes.
- **Validation process:**
 - Test models' performance will also be compared with simple forecasting baselines, such as seasonal averages or prior-month ED rates, to ensure meaningful improvement.
 - Cross-check predictions against CHS respiratory prevalence rates.
- **Success criteria:**
 - MAE within 15% for at least 70% of evaluated NTAs across all five boroughs
 - Statistically significant correlation ($p < 0.05$) between the composite pollen index and ED visit rate at a 2-week or 4-week lag
 - Environmental features, including pollen-related indicators, contribute to the model's predictive performance, as indicated by feature importance analysis.